EDUX / ACADEMIC INTELLIGENCE

FULL EXAM ANALYSIS REPORT

QA Three-Subject Fixture — extensive revision priorities and mathematical solutions

JEE Main 2027 · Full Syllabus Mock · JEE Main (Paper 1) · All Subjects · Generated 02 Nov 2026

STUDENT	ROLL NUMBER	EXAMINATION DATE	TOTAL MARKS
Aarav Sharma	21CS114	2026-08-01	360

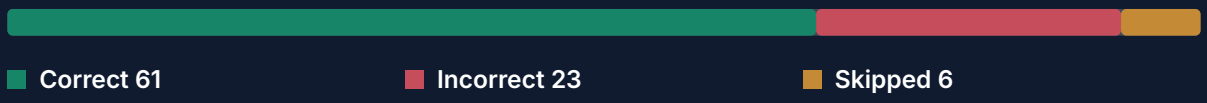
TOTAL MARKS OBTAINED

244

/ 360 marks

61 correct | 23 incorrect | 6 skipped





Correct 61 · Incorrect 23 · Skipped 6

SCORE	ACCURACY	RESPONSE DISTRIBUTION
67.8%	72.6%	61 correct | 23 incorrect | 6 skipped

PUBLISHED PERFORMANCE BRIEF

Synthetic three-subject fixture with a deliberately large published revision list and long mathematical explanations. It is not a student report.

FAR / QA-LARGE-2026-03 · QA-LARGE-2026-03 · Source fingerprint EDD82655 · 2026-11-02T09:00:00.000Z

01 PERFORMANCE INTELLIGENCE

Performance intelligence

Response composition

90 questions in the paper



- Correct 61
- Incorrect 23
- Skipped 6

Difficulty analysis

Questions and published accuracy

Easy		30 Q · 82%
Medium		30 Q · 66%
Hard		15 Q · 41%

Accuracy values are reproduced from the payload.

Mistake distribution

Count by published classification

Concept Error	5
Calculation Error	3
Formula Error	2
NCERT Detail Error	2
Silly Mistake	1
Time Pressure	1
Guess Attempt	0

Seven source classifications; rows are measured independently.

Subject-wise performance

Accuracy by subject

Physics	76%
Chemistry	71%
Mathematics	63%

Physics 88 / 120 Chemistry 80 / 120 Mathematics 76 / 120

Important performance signals

GRADE	RANK	PERCENTILE	BATCH RANK	COHORT SIZE	OVERALL SPEED	CONFIDENCE INDEX	READINESS SCORE
A	1260	94.2	12	180	4.1	87	78

HEALTH SCORE	NEGATIVE MARKS	GUESS ATTEMPTS	ATTEMPT RATIO	TOTAL QUESTIONS	SUCCESS RATE	BADGE
82	5.25	6	90.7%	90	70.1%	Top 10% — National

02 MASTERY MAP

Subject and chapter mastery

Subject marks, accuracy and classification

SUBJECT	MARKS	ACCURACY	RANK	LEVEL	STRONG AREAS	WEAK AREAS
Physics	88 / 120	76%	6	Medium	Mechanics	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations
Chemistry	80 / 120	71%	9	Easy	Physical Chemistry	Coordination Compounds
Mathematics	76 / 120	63%	14	Hard	Algebra	Definite Integration

Strong, weak and critical chapters by subject

SUBJECT	STRONG CHAPTERS	WEAK CHAPTERS	CRITICAL CHAPTERS
Physics	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations · Electrostatics · Modern Physics	Ray Optics	—
Chemistry	Physical Chemistry	Coordination Compounds	—
Mathematics	Algebra	Calculus · Probability & Stats	Calculus · Probability & Stats

Chapter mastery · grouped by subject

CHAPTER	SUBJECT	ACCURACY	MARKS	TIME	MASTERY
Physics					
Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Physics	44%	6	4m	Strong
Electrostatics	Physics	51%	7	5m	Strong
Modern Physics	Physics	58%	8	6m	Strong
Ray Optics	Physics	65%	9	7m	Weak
Chemistry					
Physical Chemistry	Chemistry	72%	10	8m	Strong
Organic Chemistry	Chemistry	79%	11	9m	Average

CHAPTER	SUBJECT	ACCURACY	MARKS	TIME	MASTERY
Inorganic Chemistry	Chemistry	86%	12	10m	Average
Chemistry					
Coordination Compounds	Chemistry	45%	13	4m	Weak
Mathematics					
Algebra	Mathematics	52%	14	5m	Strong
Coordinate Geometry	Mathematics	59%	15	6m	Average
Calculus	Mathematics	66%	16	7m	Critical
Probability & Stats	Mathematics	73%	17	8m	Critical

03 PACE INTELLIGENCE

Time intelligence

Chapter time consumption · measured rows

CHAPTER	SUBJECT	SHARE OF TOTAL TIME	TIME CONSUMED	SHARE
Physics				
Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Physics	5%	4m	5%
Electrostatics	Physics	6%	5m	6%
Modern Physics	Physics	8%	6m	8%
Ray Optics	Physics	9%	7m	9%
Chemistry				
Physical Chemistry	Chemistry	10%	8m	10%
Organic Chemistry	Chemistry	11%	9m	11%
Inorganic Chemistry	Chemistry	13%	10m	13%
Coordination Compounds	Chemistry	5%	4m	5%
Mathematics				
Algebra	Mathematics	6%	5m	6%
Coordinate Geometry	Mathematics	8%	6m	8%
Calculus	Mathematics	9%	7m	9%
Probability & Stats	Mathematics	10%	8m	10%

Time-management metrics

AVERAGE / QUESTION	NAVIGATION COUNT	MANAGEMENT SCORE	FASTEST QUESTION
4.1	8	76	Q3 — Mole Concept

FASTEST TIME	SLOWEST QUESTION	SLOWEST TIME
0.8	Q58 — Definite Integration	8.9

Sectional allocation and efficiency

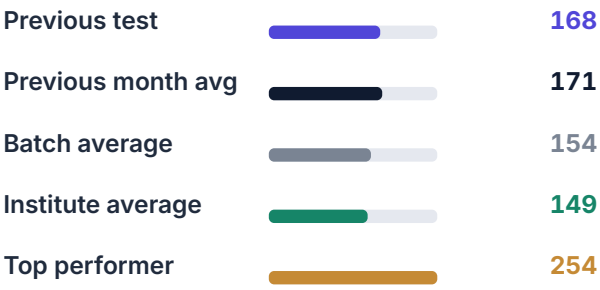
SECTION	ALLOC.	USED	EFF.
Physics	60	52	115
Chemistry	55	47	117
Mathematics	65	71	92

04 COMPARISON & TRAJECTORY

Comparison and projection

Comparative performance

Benchmark scores



Current score 244 / 360

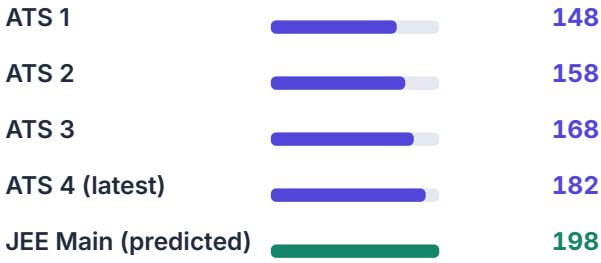
Comparison register

BENCHMARK	SCORE	PCT.	GRADE	REFERENCE
Previous test	168	87.2	B	—
Previous month avg	171	88.1	—	—
Batch average	154	82.6	—	—
Institute average	149	80.4	—	—
Top performer	254	99.6	—	Divya Krishnan

VS PREVIOUS TEST	VS PREVIOUS MONTH	VS BATCH AVERAGE	VS INSTITUTE AVERAGE
+14	+11	+28	+33

Projection and readiness

Score trajectory



Expected improvement +4.2% · Risk Low

JEE PERCENTILE	NEET SCORE	EXPECTED AIR
96.2	—	11800

TARGET PROBABILITY	RISK LEVEL	EXPECTED IMPROVEMENT
84	Low	+4.2%

Trajectory values

EXAMINATION / PROJECTION	SCORE
ATS 1	148
ATS 2	158
ATS 3	168
ATS 4 (latest)	182
JEE Main (predicted)	198

READINESS SCORE	CONFIDENCE INDEX	HEALTH SCORE	OVERALL ACCURACY	SUCCESS RATE	BADGE
78	87	82	73.9%	70.1%	Top 10% — National

05 QUESTION INTELLIGENCE

Question-by-question analysis

14 question-review rows supplied. Missing optional values are shown as —; full source explanations are never shortened.

REVIEW LEGEND

Correct · Incorrect · Skipped Each question presents its identity, classifications, answer comparison, marks, time, mistake category and complete solution.

Q1

Q1 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Physics	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Mechanics

DIFFICULTY	TYPE	STATUS
—	MCQ	Correct

QUESTION / OPTIONS

Question: Derive case 1: $\frac{d}{dx}\left(x^2\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{3} k^2$. Options: A · $\frac{\pi}{4}$ B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the question area

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
A	B	4	2m 06s

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

SOLUTION / HINT

Check the sign convention and the limits of integration carefully.

MISTAKE NOTE

Used u = x^2 instead of u = x^2+1

Q2

Q2 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Physics	—	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions

DIFFICULTY	TYPE	STATUS
—	MCQ	Correct

QUESTION / OPTIONS

Question: Derive case 2: $\frac{d}{dx}\left(x^3\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{4} k^2$. Options: A · $\frac{\pi}{4}$ B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the question area

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
B	C	4	2m 48s

Mistake category · Concept Error

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

Q3 Q3 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	—	Mole Concept
DIFFICULTY	**TYPE**	**STATUS**
—	Numerical	Correct

QUESTION / OPTIONS

Question: Derive case 3: $\frac{d}{dx}\left(x^4\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{5} k^2$. Options: A · $\frac{\pi}{4}$ B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the question area

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
C	D	4	48s

EXPLANATION

A multi-line derivation:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}
```

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

Q5 Q5 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Chemistry	—	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions
DIFFICULTY	**TYPE**	**STATUS**
—	Assertion Reason	Incorrect

QUESTION / OPTIONS

Question: Derive case 4: $\frac{d}{dx}\left(x^5\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{6} k^2$. Options: A · $\frac{\pi}{4}$ B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the question area

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
D	A	0	3m 12s

EXPLANATION

FLAGGED SOURCE — do not correct:

Q5 · solution continued

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
\frac{1}{2

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
\int_0^1 \frac{x}{\sqrt{1-x^2}

. Neither fragment is guessed or silently repaired.

SOLUTION / HINT

Check the sign convention and the limits of integration carefully.

Q7

Q7 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Mathematics	—	Algebra
DIFFICULTY	**TYPE**	**STATUS**
—	MCQ	Correct

QUESTION / OPTIONS

Question: Derive case 5: $\frac{d}{dx}\left(x^6\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{7} k^2$. Options: A · $\frac{\pi}{4}$ B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the question area

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
A	B	4	3m

Mistake category · Concept Error

SOLUTION / EXPLANATION

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

Q14

Q14 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Mathematics	—	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions

DIFFICULTY	TYPE	STATUS
—	MCQ	Incorrect

QUESTION / OPTIONS

Question: Derive case 6: $\frac{d}{dx}\left(x^7\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{8} k^2$. Options: A · $\frac{\pi}{4}$ B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the question area

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
B	C	0	5m 36s

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

MISTAKE NOTE

Used u = x^2 instead of u = x^2+1

Q18

Q18 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC
Physics	—	Ray Optics
DIFFICULTY	**TYPE**	**STATUS**
—	MCQ	Skipped

QUESTION / OPTIONS

Question: Derive case 7: $\frac{d}{dx}\left(x^{8}\sqrt{1-x^{2}}\right)$ and evaluate $\sum_{k=1}^{9} k^{2}$. Options: A · $\frac{\pi}{4}$ B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the question area

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
C	D	0	4m 24s

EXPLANATION

A multi-line derivation:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}
```

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

SOLUTION / HINT

Check the sign convention and the limits of integration carefully.

Q23

Q23 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	—	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions

DIFFICULTY	TYPE	STATUS
—	Integer	Incorrect

QUESTION / OPTIONS

Question: Derive case 8: $\frac{d}{dx}\left(x^9\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{10} k^2$. Options: A · $\frac{\pi}{4}$ B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the question area

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
D	A	0	4m 54s

Mistake category · Concept Error

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
\frac{1}{2

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
\int_0^1 \frac{x}{\sqrt{1-x^2}

. Neither fragment is guessed or silently repaired.

Q31 Q31 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Chemistry	—	Coordination
DIFFICULTY	**TYPE**	**STATUS**
—	MCQ	Incorrect

QUESTION / OPTIONS

Question: Derive case 9: $\frac{d}{dx}\left(x^{10}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{11} k^2$. Options: A · $\frac{\pi}{4}$ B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the question area

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
A	B	0	3m 48s

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

Q44 Q44 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Mathematics	—	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions
DIFFICULTY	**TYPE**	**STATUS**
—	MCQ	Correct

QUESTION / OPTIONS

Question: Derive case 10: $\frac{d}{dx}\left(x^{11}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{12} k^2$. Options: A · $\frac{\pi}{4}$ B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the question area

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
B	C	4	4m

EXPLANATION

Using

Q44 · solution continued

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

SOLUTION / HINT

Check the sign convention and the limits of integration carefully.

Q47 Q47 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Mathematics	—	3D Geometry
DIFFICULTY	TYPE	STATUS
—	Numerical	Incorrect

QUESTION / OPTIONS

Question: Derive case 11: $\frac{d}{dx}\left(x^{12}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{13} k^2$. Options: A · $\frac{\pi}{4}$ B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the question area

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
C	D	0	5m 24s

Mistake category · Concept Error

SOLUTION / EXPLANATION

EXPLANATION

A multi-line derivation:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}
```

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

MISTAKE NOTE

Used u = x^2 instead of u = x^2+1

Q58

Q58 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC
Mathematics	—	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions

DIFFICULTY	TYPE	STATUS
—	MCQ	Skipped

QUESTION / OPTIONS

Question: Derive case 12: $\frac{d}{dx}\left(x^{13}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{14} k^2$. Options: A · $\frac{\pi}{4}$ B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the question area

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
D	A	0	8m 54s

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
\frac{1}{2

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
\int_0^1 \frac{x}{\sqrt{1-x^2}

. Neither fragment is guessed or silently repaired.

Q63

Q63 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Mathematics	—	Probability
DIFFICULTY	**TYPE**	**STATUS**
—	MCQ	Incorrect

QUESTION / OPTIONS

Question: Derive case 13: $\frac{d}{dx}\left(x^{14}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{15} k^2$. Options: A · $\frac{\pi}{4}$ B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the question area

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
A	B	0	4m 12s

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

SOLUTION / HINT

Check the sign convention and the limits of integration carefully.

Q71

Q71 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Chemistry	—	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions

DIFFICULTY	TYPE	STATUS
—	MCQ	Incorrect

QUESTION / OPTIONS

Question: Derive case 14: $\frac{d}{dx}\left(x^{15}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{16} k^2$. Options: A · $\frac{\pi}{4}$ B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the question area

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
B	C	0	3m 36s

Mistake category · Concept Error

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

06 LEARNING INTELLIGENCE

Learning intelligence

Published revision priorities

Priority revision register

PRIORITY TOPIC	CLASSIFICATION	TIMEFRAME
Critical priority topic 1 — Definite Integration — properties & substitution	Critical	48 hours
High priority topic 2 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	High	5 days
Medium priority topic 3 — Definite Integration — properties & substitution	Medium	7 days
Critical priority topic 4 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Critical	10 days
High priority topic 5 — Definite Integration — properties & substitution	High	48 hours
Medium priority topic 6 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Medium	5 days
Critical priority topic 7 — Definite Integration — properties & substitution	Critical	7 days
High priority topic 8 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	High	10 days
Medium priority topic 9 — Definite Integration — properties & substitution	Medium	48 hours
Critical priority topic 10 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Critical	5 days
High priority topic 11 — Definite Integration — properties & substitution	High	7 days
Medium priority topic 12 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Medium	10 days

Topic-level accuracy

TOPIC	SUBJECT	ACC.	LEVEL
Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Physics	38%	Critical
Work-Energy-Power	Physics	47%	Weak
Electrostatics — Gauss Law	Physics	56%	Average
Thermodynamics (Physics)	Physics	65%	Strong
Ray Optics — Lenses	Physics	74%	Critical
Semiconductors	Physics	83%	Weak
Mole Concept	Chemistry	92%	Average
Chemical Bonding	Chemistry	43%	Strong
Coordination Compounds	Chemistry	52%	Critical
Definite Integration	Mathematics	61%	Weak
3D Geometry	Mathematics	70%	Average
Probability	Mathematics	79%	Strong
Quadratic Equations	Mathematics	88%	Critical
Matrices & Determinants	Mathematics	39%	Weak

Mistake evidence

Q	SOURCE DETAIL
Q14	Misapplied substitution — used $u = x^2$ instead of $u = x^2+1$
Q23	Decimal error in $1/r^2$ computation
Q31	Confused magnetic moment values for d^4 vs d^5
Q47	Wrong distance formula between skew lines
Q52	Sign convention error in lens formula
Q63	Counted 36 outcomes instead of 36 — swapped total
Q71	Rushed mechanism — skipped a step

Weak chapters

NO.	WEAK CHAPTER
1	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations
2	Calculus (48%)
3	Probability & Stats (45%)
4	Coordination Compounds (52%)
5	Ray Optics (55%)
6	Organic Reaction Mechanisms (49%)

Subject-wise revision sections

Priority topics grouped by subject

SUBJECT	PRIORITY TOPIC	CHAPTER	ACCURACY	CLASSIFICATION
Physics				
Physics	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	—	38%	Critical
Physics	Work-Energy-Power	—	47%	Weak
Physics	Electrostatics — Gauss Law	—	56%	Weak
Mathematics				
Mathematics	Matrices & Determinants	—	39%	Critical
Mathematics	Definite Integration	—	61%	Review

SUBJECT	PRIORITY TOPIC	CHAPTER	ACCURACY	CLASSIFICATION
Mathematics	3D Geometry	—	—	Revise
Chemistry				
Chemistry	Chemical Bonding	—	43%	Critical
Chemistry	Coordination Compounds	—	52%	Weak
Also revise				
Also revise	Ray Optics sign convention	—	—	Revise
Also revise	Coordination Compounds — NCERT tables	—	—	Revise
Also revise	Probability — Bayes & distributions	—	—	Revise
Also revise	Thermodynamic non-ideal corrections	—	—	Revise
Also revise	Electrostatic boundary conditions	—	—	Revise

Recommended next actions

Published action register

ACTION	RESOURCE	DETAIL
PYQ	JEE Main 2024 — Definite Integration (5 Qs)	PYQ Bank
PYQ	JEE Main 2023 — Ray Optics (4 Qs)	PYQ Bank
PYQ	JEE Advanced 2022 — Coordination Compounds	PYQ Bank
Practice	20-question drill — Integration by substitution	20 · Medium
Practice	10-question drill — Lens formula applications	10 · Medium
Mock test	Sectional mock — Mathematics (60 min)	Sectional
Mock test	Full syllabus mock — JEE Main pattern	Full
Lecture	Integral Calculus — Masterclass (45 min)	Video
Lecture	Ray Optics — Sign Convention Simplified	Video

Weak topics / published list

NO.	WEAK TOPIC
1	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions
2	Definite Integration
3	3D Geometry
4	Ray Optics sign convention
5	Coordination Compounds — NCERT tables
6	Probability — Bayes & distributions
7	Thermodynamic non-ideal corrections
8	Electrostatic boundary conditions

PUBLISHED PERFORMANCE BRIEF

Synthetic three-subject fixture with a deliberately large published revision list and long mathematical explanations. It is not a student report.

08 DATA INTEGRITY

Source data register

Source data register · 286 canonical payload values

SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE
meta.examId QA-LARGE-2026-03	meta.course JEE Main 2027 · Full Syllabus Mock	meta.pattern JEE Main (Paper 1)	meta.student Aarav Sharma
meta.rollNo 21CS114	meta.date 2026-08-01	meta.totalMarks 360	meta.duration 3 hrs
meta.sections[0] Physics	meta.sections[1] Chemistry	meta.sections[2] Mathematics	hero.score 244
hero.percentage 67.8	hero.grade A	hero.rank 1260	hero.percentile 94.2
hero.batchRank 12	hero.cohortSize 180	hero.overallAccuracy 73.9	hero.overallSpeed 4.1
hero.confidenceIndex 87	hero.readinessScore 78	hero.healthScore 82	hero.badge Top 10% — National
questionIntelligence.total 90	questionIntelligence.attempted 84	questionIntelligence.correct 61	questionIntelligence.incorrect 23
questionIntelligence.skipped 6	questionIntelligence.negativeMarks 5.25	questionIntelligence.accuracy 72.6	questionIntelligence.successRate 70.1
questionIntelligence.attemptRatio 90.7	subjects[0].score 88	subjects[0].maxMarks 120	subjects[0].accuracy 76
subjects[0].time 52	subjects[0].difficulty Medium	subjects[0].attempted 28	subjects[0].correct 21
subjects[0].strongAreas[0] Mechanics	subjects[1].score 80	subjects[1].accuracy 71	subjects[1].time 48
subjects[1].rank 9	subjects[1].difficulty Easy	subjects[1].attempted 29	subjects[1].weakAreas[0] Coordination Compounds

SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE
subjects[1].strongAreas[0]	subjects[2].accuracy	subjects[2].rank	subjects[2].difficulty
Physical Chemistry	63	14	Hard
subjects[2].attempted	subjects[2].correct	subjects[2].weakAreas[0]	subjects[2].strongAreas[0]
27	19	Definite Integration	Algebra
chapters[0].accuracy	chapters[0].time	chapters[0].attempted	chapters[0].mastery
44	4	95	Strong
chapters[1].chapter	chapters[1].accuracy	chapters[1].marks	chapters[1].time
Electrostatics	51	7	5
chapters[1].attempted	chapters[2].chapter	chapters[2].accuracy	chapters[2].marks
92	Modern Physics	58	8
chapters[3].chapter	chapters[3].accuracy	chapters[3].mastery	chapters[4].accuracy
Ray Optics	65	Weak	72
chapters[4].marks	chapters[4].attempted	chapters[5].chapter	chapters[5].accuracy
10	96	Organic Chemistry	79
chapters[5].marks	chapters[5].mastery	chapters[6].chapter	chapters[6].accuracy
11	Average	Inorganic Chemistry	86
chapters[6].attempted	chapters[7].accuracy	chapters[7].marks	chapters[8].attempted
85	45	13	94
chapters[9].chapter	chapters[9].accuracy	chapters[9].marks	chapters[10].chapter
Coordinate Geometry	59	15	Calculus
chapters[10].accuracy	chapters[10].marks	chapters[10].mastery	chapters[11].chapter
66	16	Critical	Probability & Stats
chapters[11].accuracy	chapters[11].marks	chapters[11].attempted	topics[0].mastery
73	17	68	38
topics[1].topic	topics[1].mastery	topics[2].topic	topics[2].mastery
Work-Energy-Power	47	Electrostatics — Gauss Law	56
topics[3].topic	topics[4].topic	topics[4].mastery	topics[5].topic
Thermodynamics (Physics)	Ray Optics — Lenses	74	Semiconductors
topics[5].mastery	topics[6].topic	topics[7].topic	topics[7].mastery
83	Mole Concept	Chemical Bonding	43
topics[10].topic	topics[10].mastery	topics[11].topic	topics[12].topic
3D Geometry	70	Probability	Quadratic Equations
topics[13].topic	topics[13].mastery	mistakes[0].category	mistakes[1].category
Matrices & Determinants	39	Concept Error	Calculation Error
mistakes[1].count	mistakes[2].category	mistakes[2].count	mistakes[3].category
3	Formula Error	2	NCERT Detail Error
mistakes[4].category	mistakes[4].count	mistakes[5].category	mistakes[6].category
Silly Mistake	1	Time Pressure	Guess Attempt
mistakes[6].count	mistakeList[0].q	mistakeList[0].detail	mistakeList[1].q
0	Q14	Misapplied substitution — used u = x^2 instead of u = x^2+1	Q23
mistakeList[1].detail	mistakeList[2].q	mistakeList[2].detail	mistakeList[3].q
Decimal error in $1/r^2$ computation	Q31	Confused magnetic moment values for d^4 vs d^5	Q47
mistakeList[3].detail	mistakeList[4].q	mistakeList[4].detail	mistakeList[5].q
Wrong distance formula between skew lines	Q52	Sign convention error in lens formula	Q63
mistakeList[5].detail	mistakeList[6].q	mistakeList[6].topic	mistakeList[6].detail
Counted 36 outcomes instead of 36 — swapped total	Q71	Organic — Reactions	Rushed mechanism — skipped a step
difficulty[0].questions	difficulty[0].time	difficulty[1].time	difficulty[2].accuracy
30	3.1	4.4	41

SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE
difficulty[2].time 6.2	timeIntelligence.fastestQuestion.q Q3 — Mole Concept	timeIntelligence.fastestQuestion.time 0.8	timeIntelligence.slowestQuestion.q Q58 — Definite Integration
timeIntelligence.slowestQuestion.time 8.9	timeIntelligence.distribution[0].allocated 60	timeIntelligence.distribution[0].efficiency 115	timeIntelligence.distribution[1].allocated 55
timeIntelligence.distribution[1].efficiency 117	comparison.previousTest.label Previous test	comparison.previousTest.score 168	comparison.previousTest.percentile 87.2
comparison.previousTest.grade B	comparison.previousMonth.label Previous month avg	comparison.previousMonth.score 171	comparison.previousMonth.percentile 88.1
comparison.batchAverage.label Batch average	comparison.batchAverage.score 154	comparison.batchAverage.percentile 82.6	comparison.instituteAverage.label Institute average
comparison.instituteAverage.score 149	comparison.instituteAverage.percentile 80.4	comparison.topPerformer.label Top performer	comparison.topPerformer.score 254
comparison.topPerformer.percentile 99.6	comparison.topPerformer.name Divya Krishnan	comparison.deltas[0].label vs previous test	comparison.deltas[0].value +14
comparison.deltas[0].up true	comparison.deltas[1].label vs previous month	comparison.deltas[1].value +11	comparison.deltas[2].label vs batch average
comparison.deltas[2].value +28	comparison.deltas[3].label vs institute average	comparison.deltas[3].value +33	recommendations.weakChapters[1] Calculus (48%)
recommendations.weakChapters[2] Probability & Stats (45%)	recommendations.weakChapters[3] Coordination Compounds (52%)	recommendations.weakChapters[4] Ray Optics (55%)	recommendations.weakChapters[5] Organic Reaction Mechanisms (49%)
recommendations.weakTopics[3] Ray Optics sign convention	recommendations.weakTopics[4] Coordination Compounds — NCERT tables	recommendations.weakTopics[5] Probability — Bayes & distributions	recommendations.weakTopics[6] Thermodynamic non-ideal corrections
recommendations.weakTopics[7] Electrostatic boundary conditions	recommendations.priorityRevision[0].topic Critical priority topic 1 — Definite Integration — properties & substitution	recommendations.priorityRevision[0].timeframe 48 hours	recommendations.priorityRevision[1].priority High
recommendations.priorityRevision[1].timeframe 5 days	recommendations.priorityRevision[2].topic Medium priority topic 3 — Definite Integration — properties & substitution	recommendations.priorityRevision[2].timeframe 7 days	recommendations.priorityRevision[3].timeframe 10 days
recommendations.priorityRevision[4].topic High priority topic 5 — Definite Integration — properties & substitution	recommendations.priorityRevision[6].topic Critical priority topic 7 — Definite Integration — properties & substitution	recommendations.priorityRevision[8].topic Medium priority topic 9 — Definite Integration — properties & substitution	recommendations.priorityRevision[10].topic High priority topic 11 — Definite Integration — properties & substitution
recommendations.suggestedPYQs[0].title JEE Main 2024 — Definite Integration (5 Qs)	recommendations.suggestedPYQs[0].source PYQ Bank	recommendations.suggestedPYQs[1].title JEE Main 2023 — Ray Optics (4 Qs)	recommendations.suggestedPYQs[2].title JEE Advanced 2022 — Coordination Compounds

SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE
recommendations.practiceQuestions[0].title 20-question drill — Integration by substitution	recommendations.practiceQuestions[0].count 20	recommendations.practiceQuestions[1].title 10-question drill — Lens formula applications	recommendations.mockTests[0].title Sectional mock — Mathematics (60 min)
recommendations.mockTests[0].mode Sectional	recommendations.mockTests[1].title Full syllabus mock — JEE Main pattern	recommendations.mockTests[1].mode Full	recommendations.lectures[0].title Integral Calculus — Masterclass (45 min)
recommendations.lectures[0].type Video	recommendations.lectures[1].title Ray Optics — Sign Convention Simplified	prediction.jeePercentile 96.2	prediction.expectedAIR 11800
prediction.riskLevel Low	prediction.expectedImprovement +4.2%	prediction.trajectory[0].exam ATS 1	prediction.trajectory[0].score 148
prediction.trajectory[1].exam ATS 2	prediction.trajectory[1].score 158	prediction.trajectory[2].exam ATS 3	prediction.trajectory[3].exam ATS 4 (latest)
prediction.trajectory[3].score 182	prediction.trajectory[4].exam JEE Main (predicted)	prediction.trajectory[4].score 198	questionReview[0].q Q1
questionReview[0].type MCQ	questionReview[0].status Correct	questionReview[0].time 2.1	questionReview[0].options[0] $$\frac{\pi}{4}$$
questionReview[0].options[1] $\sqrt{3}$ s	questionReview[0].options[2] 4 s	questionReview[0].options[3] A very long alternative answer that must wrap without clipping the question area	questionReview[0].hint Check the sign convention and the limits of integration carefully.
questionReview[0].mistakeHint Used u = x^2 instead of u = x^2+1	questionReview[1].q Q2	questionReview[1].time 2.8	questionReview[2].q Q3
questionReview[2].type Numerical	questionReview[3].q Q5	questionReview[3].type Assertion Reason	questionReview[3].status Incorrect
questionReview[3].time 3.2	questionReview[4].q Q7	questionReview[5].time 5.6	questionReview[6].q Q18
questionReview[6].status Skipped	questionReview[7].type Integer	questionReview[7].time 4.9	questionReview[8].topic Coordination
questionReview[8].time 3.8	questionReview[9].q Q44	questionReview[10].time 5.4	questionReview[11].q Q58
questionReview[12].time 4.2	questionReview[13].time 3.6	— —	— —

Extended source values · 28 verbatim entries

SOURCE / VALUE	SOURCE / VALUE
meta.examName QA Three-Subject Fixture — extensive revision priorities and mathematical solutions	hero.aiSummary Synthetic three-subject fixture with a deliberately large published revision list and long mathematical explanations. It is not a student report.
subjects[0].weakAreas[0] Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	topics[0].topic Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions
recommendations.priorityRevision[1].topic High priority topic 2 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	recommendations.priorityRevision[3].topic Critical priority topic 4 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions

SOURCE / VALUE	SOURCE / VALUE
recommendations.priorityRevision[5].topic Medium priority topic 6 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	recommendations.priorityRevision[7].topic High priority topic 8 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions
recommendations.priorityRevision[9].topic Critical priority topic 10 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	recommendations.priorityRevision[11].topic Medium priority topic 12 — Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions
questionReview[0].text Derive case 1: $\frac{d}{dx}\left(x^{2}\sqrt{1-x^2}\right)$ and evaluate $\sum_{{k=1}}^{3} k^2$.	questionReview[0].explanation The integral reduces directly: $$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$ The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.
questionReview[1].text Derive case 2: $\frac{d}{dx}\left(x^{3}\sqrt{1-x^2}\right)$ and evaluate $\sum_{{k=1}}^{4} k^2$.	questionReview[1].explanation Using $$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$ and noting $\alpha^2 + \beta^2 \leq \sqrt{\gamma}$, the solution follows. Charge balance for $\ce{H2SO4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.
questionReview[2].text Derive case 3: $\frac{d}{dx}\left(x^{4}\sqrt{1-x^2}\right)$ and evaluate $\sum_{{k=1}}^{5} k^2$.	questionReview[2].explanation A multi-line derivation: $$\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}$$ with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty} (1+1/n)^n = e$.
questionReview[3].text Derive case 4: $\frac{d}{dx}\left(x^{5}\sqrt{1-x^2}\right)$ and evaluate $\sum_{{k=1}}^{6} k^2$.	questionReview[3].explanation FLAGGED SOURCE — do not correct: $\frac{1}{2$ is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced: $$\int_0^1 \frac{x}{\sqrt{1-x^2}$$. Neither fragment is guessed or silently repaired.
questionReview[4].text Derive case 5: $\frac{d}{dx}\left(x^{6}\sqrt{1-x^2}\right)$ and evaluate $\sum_{{k=1}}^{7} k^2$.	questionReview[5].text Derive case 6: $\frac{d}{dx}\left(x^{7}\sqrt{1-x^2}\right)$ and evaluate $\sum_{{k=1}}^{8} k^2$.
questionReview[6].text Derive case 7: $\frac{d}{dx}\left(x^{8}\sqrt{1-x^2}\right)$ and evaluate $\sum_{{k=1}}^{9} k^2$.	questionReview[7].text Derive case 8: $\frac{d}{dx}\left(x^{9}\sqrt{1-x^2}\right)$ and evaluate $\sum_{{k=1}}^{10} k^2$.
questionReview[8].text Derive case 9: $\frac{d}{dx}\left(x^{10}\sqrt{1-x^2}\right)$ and evaluate $\sum_{{k=1}}^{11} k^2$.	questionReview[9].text Derive case 10: $\frac{d}{dx}\left(x^{11}\sqrt{1-x^2}\right)$ and evaluate $\sum_{{k=1}}^{12} k^2$.
questionReview[10].text Derive case 11: $\frac{d}{dx}\left(x^{12}\sqrt{1-x^2}\right)$ and evaluate $\sum_{{k=1}}^{13} k^2$.	questionReview[11].text Derive case 12: $\frac{d}{dx}\left(x^{13}\sqrt{1-x^2}\right)$ and evaluate $\sum_{{k=1}}^{14} k^2$.
questionReview[12].text Derive case 13: $\frac{d}{dx}\left(x^{14}\sqrt{1-x^2}\right)$ and evaluate $\sum_{{k=1}}^{15} k^2$.	questionReview[13].text Derive case 14: $\frac{d}{dx}\left(x^{15}\sqrt{1-x^2}\right)$ and evaluate $\sum_{{k=1}}^{16} k^2$.

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End of report · FAR / QA-LARGE-2026-03

Missing values are shown as —.